

Course Level: Undergraduate Open Elective (B. Tech)

BL: 2 Understand

CO: Explain the fundamental principles of nutrition.

Module 2: Digestion & Absorption

1. Food Complexity & Digestion Overview

Digestion involves six activities:

- Ingestion
- Mechanical/physical digestion
- Propulsion
- Chemical digestion
- Absorption
- Defecation

2. First Step: Mouth

Chewing (mastication) → mechanical digestion.

Saliva functions:

Softens and moistens food for swallowing.

Contains enzymes:

Alpha-amylase → breaks down carbohydrates.

Lingual lipase → breaks down lipids.

Protects teeth, gums, and reduces infections.

Salivary glands:

Parotid (sides of face)

Submandibular (under jaw)

Sublingual (under tongue)

Taste perception: sour, sweet, salty, bitter, savoury.

Other sensations: spiciness, coolness (via nerves).

Teeth roles:

Incisors → cutting/gnawing.

Molars → grinding/chewing.

Saliva pH: 6.2–7.6; bacteria lower pH by producing acids.

Food moves to oesophagus via peristalsis.

3. Stomach

Food + salivary enzymes continue digestion.

Peristalsis moves food within stomach.

Stomach secretes digestive juices & enzymes.

pH changes:

Normal: 5.0–6.0

After food: drops below 3.0 (within ~45 min).

Acid activates enzymes (does not digest food directly).

4. Intestine

Small intestine:

Duodenum → breakdown.

Jejunum & Ileum → absorption.

Peristalsis continues.

Accessory organs (pancreas, liver, gall bladder) contribute enzymes/bile.

Large intestine:

Processes waste → stool formation.

Bacteria synthesize vitamins, process waste, protect against harmful microbes.

Transit times:

Stomach + small intestine: 6–8 hrs.

Colon: ~36 hrs.

Entire GI tract: 2–5 days.

5. Accessory Organs

Pancreas: enzymes for fats, proteins, carbs.

Liver: produces bile, purifies blood.

Gall bladder: stores bile; aids fat absorption & waste removal.

6. Digestive Enzymes

Main enzymes:

Amylase → carbs → sugars.

Lipase → fats → fatty acids + glycerol.

Protease → proteins → amino acids.

Other enzymes:

Maltase → maltose → glucose.

Lactase → lactose → glucose + galactose.

Sucrase → sucrose → glucose + fructose.

Sources: salivary glands, stomach, pancreas, small intestine.

Natural enzyme-rich foods: papaya, mango, kiwi, banana, honey, ginger, avocado.

7. Food Factors Affecting Gastric Residence Time

Consistency: liquids leave faster (water ~10 min for 50% emptying).

Calorie content: low-calorie foods exit faster.

Nutrient type: carbs/proteins digest faster; fats/fibre stay longer.

Volume